Education

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Poll

Evidence suggests you're not learning anything useful for your future earnings. College is just a fancy way to signal that you are smarter than people who could not get into Berkeley.

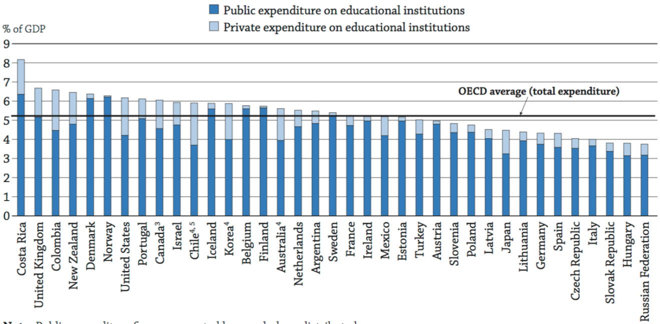
- A. True
- B. False
- C. Uncertain

Education

- Education is one of the 3 largest programs funded by government (along with retirement and health)
- All advanced economies fund most (80% on average) of education (pre-K, K-12, higher ed) through government
- Education level highly dependent on govt policy (see OECD stats)
- In US, 4.5% of GDP or 1/7 of total government expenditure
- In US, 80% of education spending done at the state and local level
- Focus of an extensive body of research in the rapidly expanding field of economics of education

Figure B2.1. Public and private expenditure on educational institutions, as a percentage of GDP (2013)

From public[1] and private[2] sources



Note: Public expenditure figures presented here exclude undistributed programme.

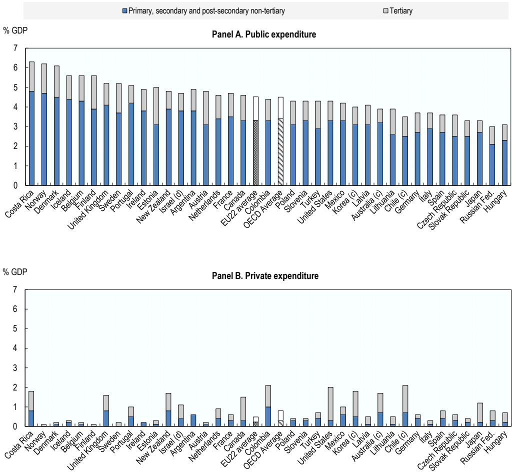
1. Including public subsidies to households attributable to educational institutions, and direct expenditure on educational institutions from international sources.
2. Net of public subsidies attributable for educational institutions.
3. Year of reference 2012.
4. Public does not include international sources.
5. Year of reference 2014.

Countries are ranked in descending order of expenditure from both public and private sources on educational institutions.

Source: OECD. Table B2.3. See Annex 3 for notes (www.oecd.org/education/education-at-a-glance-19991487.htm).

Chart PF1.2.A Expenditure on education as % of GDP, by level of education and source of funds, 2013[a]

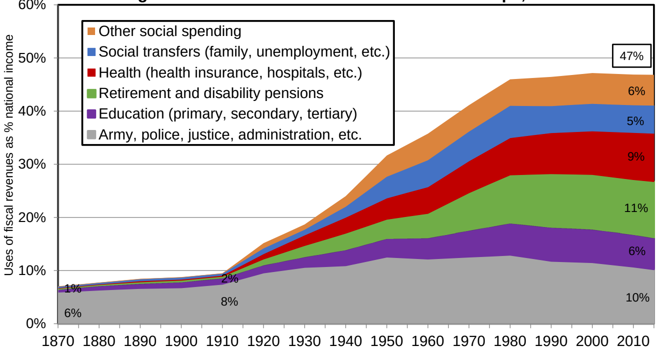
Expenditure on primary, secondary and post-secondary non-tertiary and on tertiary education by public or private source[b], as % of GDP



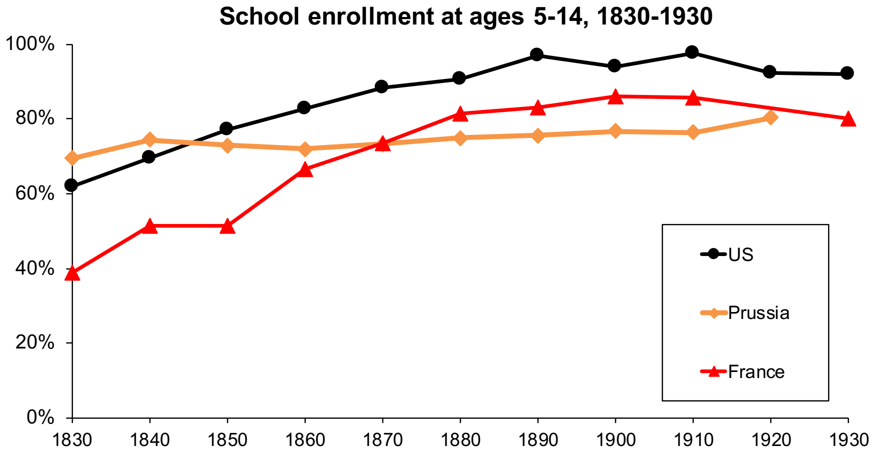
Why Should the Government Be Involved in Education?

- Education is long and costly (teachers+schools, US cost is $15K/year-kid) and everybody needs it in modern economy
- Key reason among the public: Without govt provision, low income families would not be able to afford it for their kids (would hurt opportunity)
- Governments developed mass education in 20th century [mandatory up to certain ages and publicly provided]
- Played a big role in fostering economic development:
  1. Empowers women and fertility transition in developing countries
  2. Modern economy requires an educated workforce

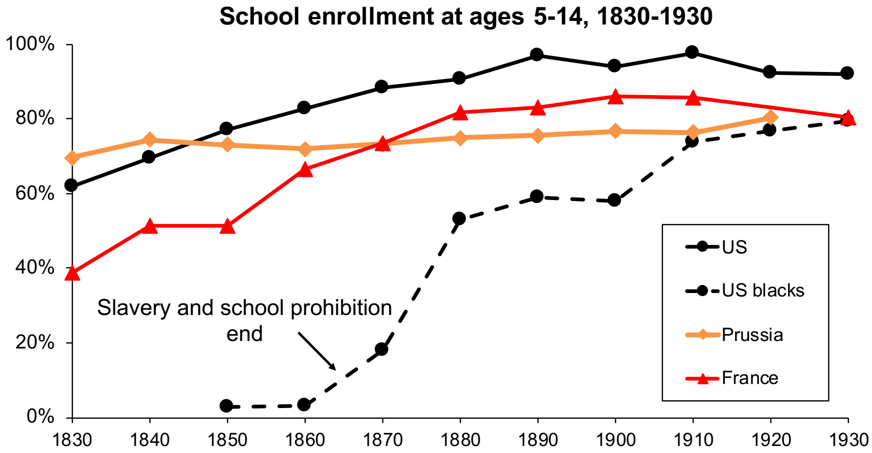
Figure 10.15. The rise of the social State in Europe, 1870-2015



Interpretation. In 2015, fiscal revenues represented 47% of national income on average in Western Europe et were used as follows: 10% of national income for regalian expenditure (army, police, justice, general administration, basic infrastructure: roads, etc.); 6% for education; 11% for pensions; 9% for health; 5% for social transfers (other than pensions); 6% for other social spending (housing, etc.). Before 1914, regalian expenditure absorbed almost all fiscal revenues. **Note.** The evolution depicted here is the average of Germany, France, Britain and Sweden (see figure 10.14). Sources and séries: see piketty.pse.ens.fr/ideology.



Fraction of children aged 5-14 enrolled in school (public or private).
Sources and series: Lindert (2004) Growing Public and Historical Statistics of the US



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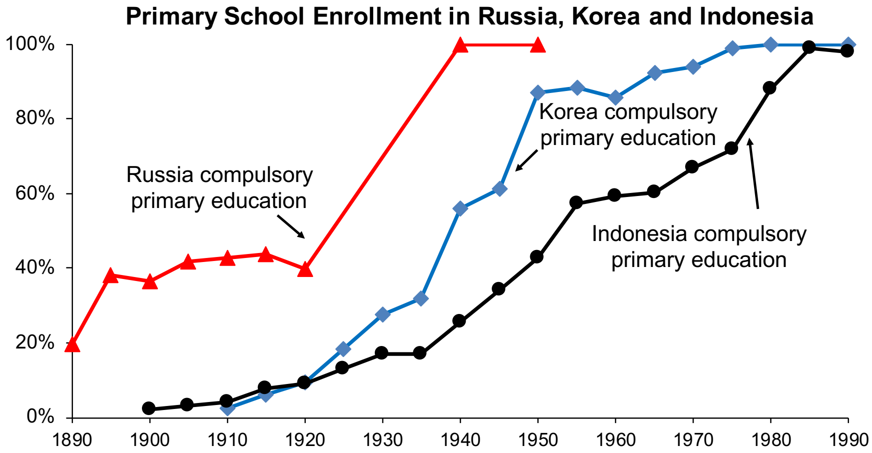


Fraction of children enrolled in primary school (public or private).
Source: Lee and Lee (2016).

Why Should the Government Be Involved in Education?

- For economists, ex-ante not obvious because education does not look like a public good
  1. Returns to education are largely private
  2. Education is excludable
- We should expect students to invest roughly the optimal amount in their own education

Why Should the Government Be Involved in Education?

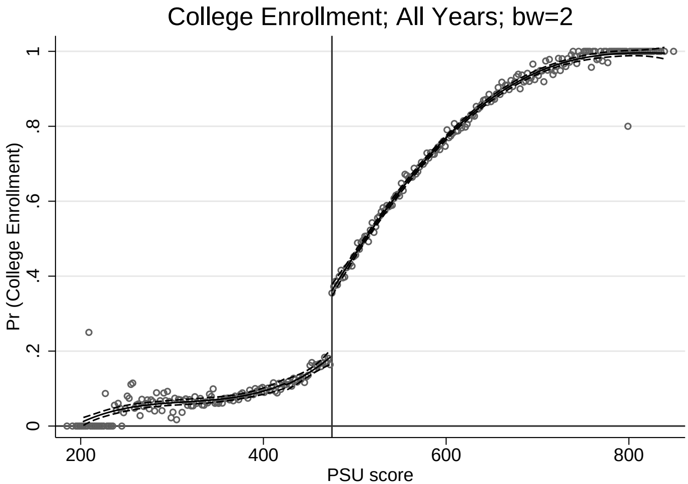
- Traditional motives pointed out by economists:
    1. Externalities (productivity spillovers, crime, citizenship)
    2. Borrowing constraints (poor but talented students may not be able to borrow against future earnings to get an education)
    3. Family and individual failures (to conform to standard econ model):
        - Some parents may not take good care of their children (public education provides opportunity for all)
        - Young adults might not do what is in their long-run interest due to self-control problems or lack of information

Externalities of Education on Crime

- Standard observational regression comparing the educated vs. not-educated:
  $\text{Crime}_i = \alpha + \beta\ \text{Educ}_i + \varepsilon_i$
- Observational regression likely biased because propensity to commit crime ε_i is likely negatively correlated with Educ_i.
- Lochner and Moretti (2004) use changes in state compulsory attendance laws as a natural experiment: State T increases compulsory attendance from 9 to 10 years at time t, State C does not.
- Can look at effect on education, and then on crime using difference-in-differences
- They show that an extra year of schooling reduces incarceration rates significantly: 0.1 pct point decline for white males relative to a mean of 1%; 0.3 pct point decline for Black males relative to mean of 3%
- Gap in schooling between white and Black accounts for more than 1/4 difference in crime rates

Borrowing Constraints: Effects of Loans

- If there are no borrowing constraints (and individuals are rational), current resources should not matter for educational decisions: invest in education only if PDV benefits > costs
- Empirical evidence shows that availability of loans do matter suggesting that borrowing constraints are an issue
- Solis (2017) studies the effects of guaranteed loans on college attendance in Chile
- Guaranteed loan is available if test score of student (equivalent of SAT for Chile) is above threshold equal to 475.
- Regression discontinuity design: discontinuity in loan availability translates into discontinuity in college attendance
- Very compelling evidence that loan availability matters



Source: Solis (2013)

Individual Failures (College Application Tutoring)

- Carrell-Sacerdote (2017) carry out a field experiment in New Hampshire high-schools
- College students from Dartmouth help senior high-schoolers to apply to college (weekly meetings in Winter semester)
- Randomization within high schools: select only 50% of seniors
- Find large positive impact on women (+15 points likelihood of enrolling in college) but small effects on men
- Also find a cash bonus for applying to colleges without tutorial does not have any impact $\Rightarrow$ Pure econonmic incentives not enough
- Effects require time intensive tutorials (that parents/teachers might otherwise provide)
- Series of papers by Roland Fryer also show that paying K-12 kids to achieve does not work (interpreted as kids not knowing how to earn higher scores without guidance)

Measuring the Returns to Education

- Returns to education: The benefits that accrue to society when students get more schooling or when they get schooling from a higher-quality environment.
- Effects of education levels on productivity
  - There is a large literature that argues that more education leads to higher wages in the labor market, beginning with the observational regressional: $\text{Earnings}_i = \alpha + \beta \cdot \text{Education}_i + \varepsilon_i$
  - There is substantial controversy, however, over the implications of this correlation ($\beta > 0$).

Effects of Education on Earnings

- 1) Education as Human Capital Accumulation
  - Human capital: A person's stock of skills, which may be increased by education
  - In that scenario, education raises earnings because it improves productivity
- 2) Education as a Screening Device
  - Screening: A model that suggests that education provides only a means of separating high-ability from low-ability individuals and does not actually improve skills.
  - In that scenario, education raises *individual* earnings but it does not improve productivity (rat-race)

Measuring the Returns to Education

- Policy Implications
  - Under the human capital model, government would want to support education or at least provide loans to individuals so that they can get more education and raise their productivity.
  - Under the screening model, however, the government would *not* want to support more education for any given individual.
- Differentiating the theories
  - Most of the returns to education reflect accumulation of human capital rather than screening
  - Example: Clark-Martorell '14 show that barely getting a high-school degree in Texas has no visible impact on later earnings

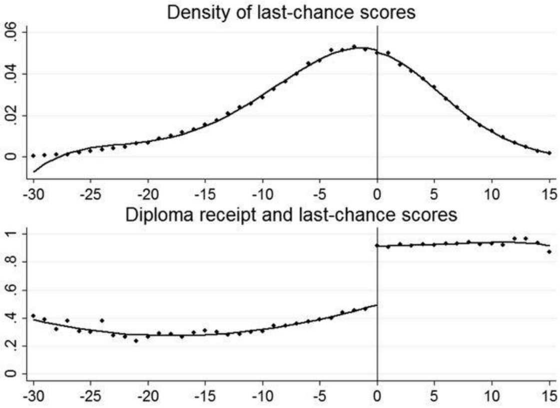


FIG. 1.—Last-chance exam scores and diploma receipt. The graphs are based on the last-chance sample. See table 1 and the text. Dots are test score cell means. The scores on the x-axis are the minimum of the section scores (recentered to be zero at the passing cutoff) that are taken in the last-chance exam. Lines are fourth-order polynomials fitted separately on either side of the passing threshold.

Source: Clark and Martorell JPE'14

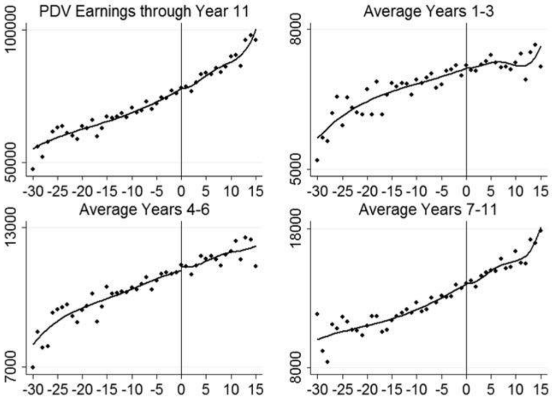


Fig. 2.—Earnings by last-chance exam scores. The graphs are based on the last-chance samples. See table 1 and the text. Dots are test score cell means. The scores on the *x*-axis are the minimum of the section scores (recentered to be zero at the passing cutoff) that are taken in the last-chance exam. Lines are fourth-order polynomials fitted separately on either side of the passing threshold.

Source: Clark and Martorell JPE'14

Evidence on the Returns to Education and Screening

- Basic observational approach: $\text{Earnings}_i = \alpha + \beta \cdot \text{Education}_i + \varepsilon_i$
- Amounts to comparing the earnings of people with different education.
- Issue: ability to earn ε_i might be correlated with education choices
- Two methods try to control for this bias in estimating the true human capital effects of education
  1. Control for underlying ability by adding variables (e.g. SAT score) in the regression so that any remaining effect of education represents true productivity effects (omitted variable bias remains a concern)
  2. Find exogenous variation in education (e.g., policy change induces more education for some group but not for another group)
- Although all of these approaches have some limitations, the result of the analysis is surprisingly consistent: each additional year of education raises wages by 7-10%

The Impact of School Quality

- A number of approaches have been taken to estimate the impact of school quality on student test scores.
- Two approaches have been used to address this issue: experimental data, and quasi-experiments using policy changes
- Findings suggest that the outcomes of efforts to improve school quality can be very dependent on the approach taken to improvements

Estimating the Effects of Class Size

- Experimental example: The state of Tennessee implemented Project STAR in 1985, randomly assigning 11,000 students (grades K–3) to small classes (13–17 students) or regular classes (22–25 students)
- Krueger-Whitmore 2001 shows positive effects of small class size on test scores
- Chetty-Friedman-Hilger-Saez-Schanzebach-Yagan 2011 linked students to college enrollment and adult earnings data: finds small positive effects on college enrollments and adult earnings.
- Note: kids and teachers also randomly assigned across classes: strong class effects are visible (due to teachers or peers) and they have long-term effects on college and earnings
- Quasi-experimental example: By the mid-1990s, California had the largest class sizes in the nation (29 students per class on average). The California state government in 1996 provided strong financial incentives for schools to reduce their class size to 20 students per class: not much effects on outcomes found but controversial

Estimating the Effects of Charter Schools

- Some districts have charter schools that are public but are not subject to all state regulations for schools (more flexibility to recruit teachers / adjust hours / curriculum)
- Oversubscribed charter schools use a lottery to assign admissions
- Generates randomized experiment allowing to estimate the causal effect of charter schools by comparing lottery winners and lottery losers
- Angrist, Pathak, Walters AEJ'13 carry out a comprehensive analysis of charter schools effects in Massachusetts
- Find that urban charter schools boost achievement relative to regular urban public schools, while rural charters reduce achievement relative to regular rural public schools
- Charter schools can have a positive or negative impact depending on what they do
- Most effective approach to education: focus on instruction time, pupil comportment, selective teacher hiring, and focus on traditional math and reading skills.

Poll

Evidence suggests you're not learning anything useful for your future earnings. College is just a fancy way to signal that you are smarter than people who could not get in to Berkeley.

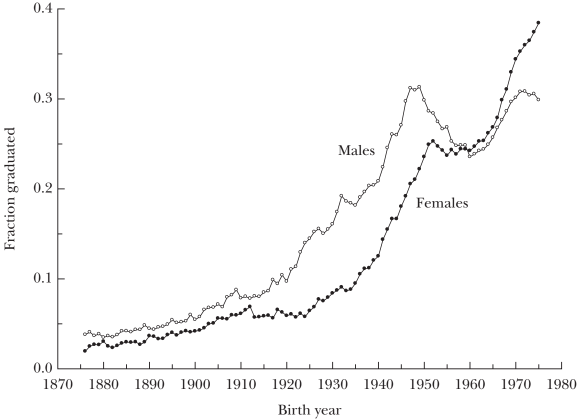
- A. True
- B. False
- C. Uncertain

Role of Government in Supply of Higher Education

- Private non-profit universities have inelastic supply (e.g., fixed student bodies at top schools such as Harvard)
- Historically, expansion of supply was carried out by public institutions (state universities and community colleges): Example: 1960 Master plan for California with 3-tier system (Community, State, University of California)
- Government push also central to increase attendance: GI Bill after WWII/Korea War increased college attendance by 15-20 points for men born 1921-1933 (Stanley QJE'03)
- Recently, states have retreated and supply has been provided by for-profit schools (get about 10% of total enrollment today)
- Deming-Goldin-Katz '12 show that for-profit schools provide little benefits, charge a lot, and are savvy at exploiting Fed Pell Grants and saddle students with debt
- Likely symptom of market failure due to individual failures/lack of information

Figure 1

College Graduation Rates (by 35 years) for Men and Women: Cohorts Born from 1876 to 1975



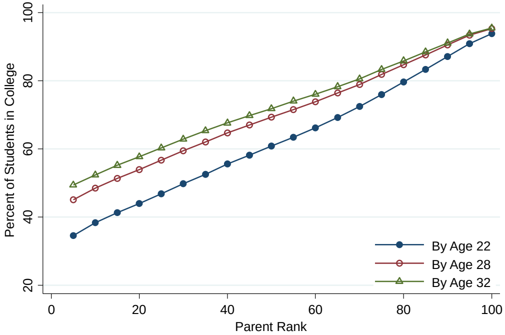
Sources: 1940 to 2000 Census of Population Integrated Public Use Micro-data Samples (IPUMS).

Role of Higher Education in Intergenerational Mobility

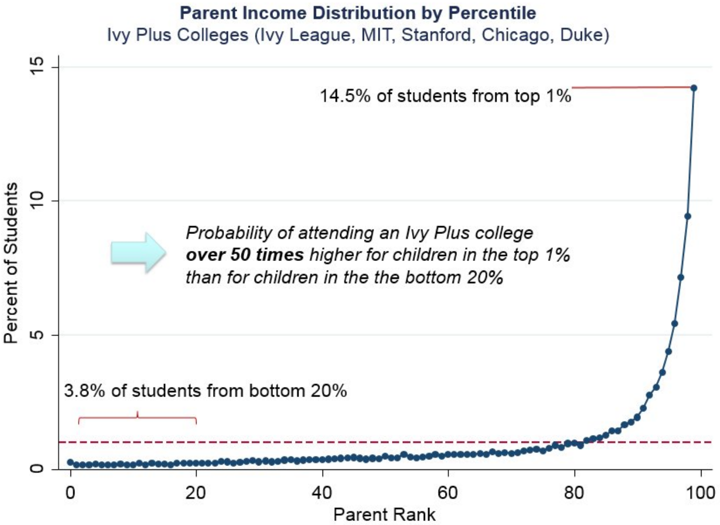
- Chetty-Friedman-Saez-Turner-Yagan '20 compile college level statistics on parental income and student earnings outcomes. Data online at [web link].
  1. Access: Huge variation in access across schools: Ivy league has more kids from top 1% families than from bottom 50%. Giving poor kids an SAT point boost in admissions (as done for legacy students) could close gap and increase intergenerational mobility.
  2. Trends: Fraction poor kids stagnated in top schools (in spite of more financial aid) and dropped at best public schools and community colleges
  3. Outcomes: Within good colleges, outcomes of poor vs. rich kids are similar $\Rightarrow$ college is the ticket to opportunity
  4. Mobility rates: Large discrepancies across colleges in fraction of students who come from bottom 20% and reach top 20% (=mobility rate)

ONLINE APPENDIX FIGURE I

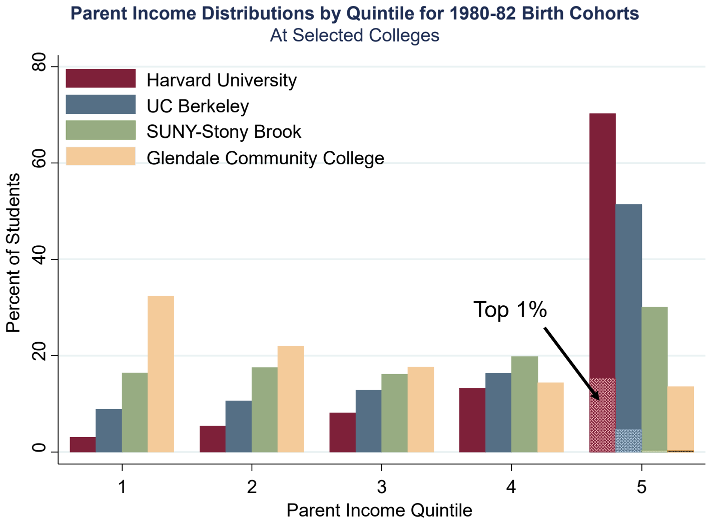
College Attendance Rates by Parent Income and Age



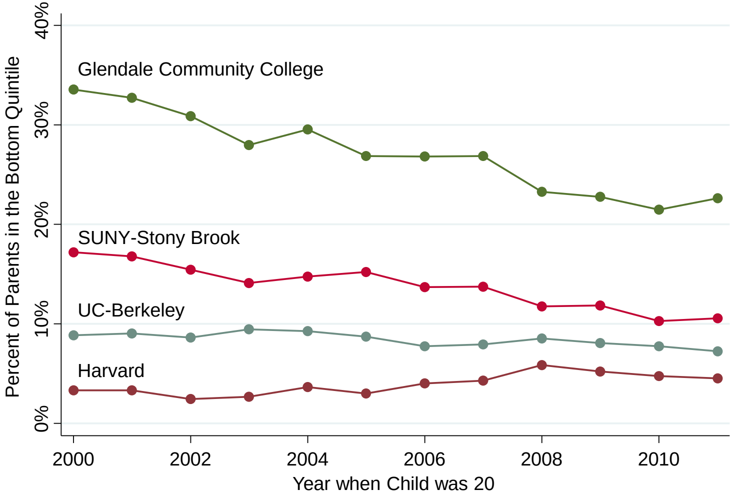
Notes: This figure plots the fraction of children in the 1980-82 birth cohorts in our analysis sample who attend college at any time during or before the year in which they turn ages 22, 28, and 32, by parent income ventile. This figure is constructed directly from the individual-level microdata.



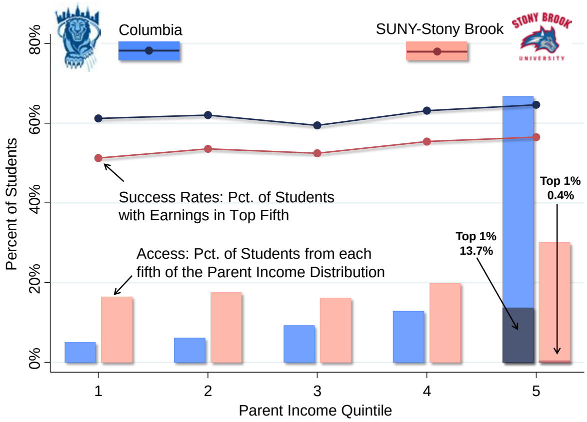
Source: Estimates from "Mobility Report Cards: The Role of Colleges in Intergenerational Mobility" (undergoing peer review) by Raj Chetty, John N. Friedman, Emmanuel Saez, Nicholas Turner, and Danny Yagan. Graph covers 1980-1982 birth cohorts. Numbers similar for 1989-1991 cohorts.


Parent Income Distributions by Quintile for 1980-82 Birth Cohorts
At Selected Colleges
Harvard University
UC Berkeley
SUNY-Stony Brook
Glendale Community College
Top 1%
Percent of Students
0
20
40
60
80
1
2
3
4
5
Parent Income Quintile

Trends in Low-Income Access from 2000-2011 at Selected Colleges



Mobility Report Cards for Columbia and SUNY-Stony Brook



Note: Bars show estimates of the fraction of parents in each quintile of the income distribution. Lines show estimates of the fraction of students from each of those quintiles who reach the top quintile as adults.

Top 10 Colleges by Mobility Rate (from Bottom to Top Quintile)

Rank	Name	Mobility Rate	=	Access	x	Success Rate
1	Cal State University – LA	9.9%		33.1%		29.9%
2	Pace University – New York	8.4%		15.2%		55.6%
3	SUNY – Stony Brook	8.4%		16.4%		51.2%
4	Technical Career Institutes	8.0%		40.3%		19.8%
5	University of Texas – Pan American	7.6%		38.7%		19.8%
6	City Univ. of New York System	7.2%		28.7%		25.2%
7	Glendale Community College	7.1%		32.4%		21.9%
8	South Texas College	6.9%		52.4%		13.2%
9	Cal State Polytechnic – Pomona	6.8%		14.9%		45.8%
10	University of Texas – El Paso	6.8%		28.0%		24.4%

Note: Table lists highest-mobility-rate colleges with more than 300 students per cohort.

Poll

What share of kids who grow up in the top 0.1% go to an Ivy Plus college (Ivy League plus MIT, Stanford, Chicago, and Duke)?

- A. 10%
- B. 25%
- C. 50%
- D. 75%
- E. 90%

Let's go to the data

Conclusion

- Education is a sector with likely substantial market failures
- Education is a sector with likely substantial impact on inequality
- Government intervenes for both reasons

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